CYCLON convergence: counterexamples and a proved special case

Research working note

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Abstract

An unrestricted local convergence claim covering the CYCLON scheduling and update conventions below is false. We prove a four-peer periodic counterexample independent of replacement conventions, and a family of strongly connected, permanently frozen multi-entry overlays under a retain-until-replaced, discard-duplicate convention. The latter works with independent uniform initiators and every shuffle length. Under mandatory partner removal, a separate availability obstruction prevents an unconditional full-capacity tuple guarantee.

We also prove eventual local uniformity for a precisely specified capacity-one retention variant with independent uniform initiators. For $n \geq 4$ its identity chain is irreducible and reversible and admits an explicit, very weak time bound. No positive convergence theorem for multi-entry, mandatory-removal CYCLON is claimed.

1 Scope and exact protocol conventions

The motivating target is a local total-variation guarantee of the kind proved for PEERSWAP in [1]. Its interchange-process reduction uses a fixed underlying graph and does not automatically extend to partial cache exchanges.

We consider a static membership $V = \{1, \ldots, n\}$, reliable atomic exchanges, and caches without self entries or duplicate identities. Descriptors contain an identity and a nonnegative integer age. Enhanced CYCLON increments the initiator's cached ages, selects an oldest descriptor as partner, sends a fresh self descriptor in place of that partner along with other selected descriptors, and receives a subset of the partner's cache [2].

The update convention matters. We distinguish:

- *Retention:* discard self and already-known incoming descriptors, and replace sent cache entries only when a usable incoming descriptor is inserted. A contacted partner is retained when no replacement is usable. Discarded duplicates do not refresh ages.
- *Mandatory removal:* unconditionally remove the contacted partner, even when the response yields no usable replacement.

These conventions differ on reciprocal exchanges. The protocol description's replacement steps and its subsequent statement that the contacted peer disappears require this edge case to be resolved. The results below state their convention explicitly; they are not claims about every implementation bearing the CYCLON name.

Time counts global activation events. Periodic phases and independent uniform random activations are treated separately. For a cache that can supply b entries, $S_i^{(b)}$ is a uniformly selected, uniformly ordered b-tuple of distinct cached identities. When a requested tuple is unavailable, write $\perp$. Uniformity at one peer and time does not imply independence across peers or times.

2 A periodic counterexample under either convention

At $c = \ell = 1$, if the initiator is p, its neighbor is q, and q's cached identity is r, with p, q, r distinct, the unique exchange outcome is

$$(f(p), f(q)) = (q, r) \quad \longmapsto \quad (f'(p), f'(q)) = (r, p). \tag{1}$$

All other identities stay fixed. Both update conventions agree here.

Theorem 1 (Failure with fixed periodic phases)**.** *There is a four-peer enhanced-CYCLON execution with a strongly connected initial overlay, full caches, $c = \ell = 1$, and one activation per peer per round, in which a peer's sample never converges to uniform. No exchange has a tie, unusable replacement, or empty cache.*

Proof. Activate peers in order $1, 2, 3, 4$ in every round, using distinct fixed phases within a common period. Initialize all descriptor ages to zero and the identity state to $(3, 4, 2, 1)$. Equation (1) gives

After activation	$f(1)$	$f(2)$	$f(3)$	$f(4)$
Initial	3	4	2	1
1	2	4	1	1
2	2	1	1	2
3	3	1	2	2
4	3	4	2	1

The initial graph is the directed cycle $1 \to 3 \to 2 \to 4 \to 1$. In every exchange p, q, r are distinct. The initiator's sole descriptor is incremented from zero to one, then discarded and replaced in its outgoing message by its own age-zero descriptor. The responder's age-zero descriptor moves to the initiator. Thus all stored ages are again zero after each exchange. The entire state and scheduler phase repeat after four activations.

At each round boundary, peer 1 returns peer 3 deterministically. For all $k \geq 0$,

$$\left\| \mathcal{L}(S_1^{(1)}(4k)) - \mathrm{Unif}\{2, 3, 4\} \right\|_{\mathrm{TV}} = \frac{2}{3}.$$

Moreover peer 1 never stores peer 4, even between round boundaries. Every mixture over observation times therefore remains at least $1/3$ from uniform. Time averaging cannot repair this execution. □

3 Frozen multi-entry states with random scheduling

This section uses retention and discards already-known descriptors *without refreshing their ages.* It allows random initiators, random subsets, and any tie-breaking rule.

Theorem 2 (Strongly connected frozen states)**.** *For every even $n \geq 4$, every $2 \leq c \leq n - 2$, and every $1 \leq \ell \leq c$, there is a strongly connected, full-cache initial state for which all cache identities remain fixed forever under enhanced CYCLON with retention. Every peer always has a unique oldest descriptor. This holds for every activation sequence and every outcome of the protocol's random subset selections.*

For every $1 \leq b \leq c$ and every observer i, its sample distance is the positive constant

$$\left\| \mathcal{L}(S_i^{(b)}(t)) - \mathrm{Unif}((V \setminus \{i\})_b) \right\|_{\mathrm{TV}} = 1 - \frac{(c)_b}{(n-1)_b}, \qquad t \geq 0, \tag{2}$$

where $(a)_b = a(a-1)\cdots(a-b+1)$.

Proof. Partition peers into pairs (p_k, q_k), $1 \le k \le m = n/2$. For each pair choose a set S_k of $c-1$ identities outside the pair that contains p_{k+1}, with indices modulo m. This is possible since $m \ge 2$ and $c - 1 \le n - 3$. Set

$$C_{p_k} = \{q_k\} \cup S_k, \qquad C_{q_k} = \{p_k\} \cup S_k.$$

Give the reciprocal partner descriptor age 1 and every other descriptor age 0. Each pair is internally mutually reachable; the edges to p_{k+1} connect the pairs around a directed ring. Consequently the overlay is strongly connected.

Suppose p_k initiates. Its unique oldest descriptor is q_k. The outgoing message consists of its own fresh identity p_k and $\ell - 1$ elements of S_k, all already in q_k's cache. The responder therefore discards every incoming descriptor. Every descriptor it returns is either p_k, which is a self entry at the initiator, or an element of S_k, already known there. The initiator also discards every incoming descriptor. No cache identity and no stored age is overwritten.

Incrementing all ages at the initiating cache preserves its strict age difference between the partner and the common neighbors. Passive responses do not increment ages. The same argument applies when q_k initiates. Induction proves that the identities remain frozen and the designated partners remain uniquely oldest for all time.

The output is uniform on the $(c)_b$ ordered tuples contained in the fixed cache, whereas the target is uniform on all $(n-1)_b$ tuples. For uniform laws on a subset A of a finite set B, their total-variation distance is $1 - |A|/|B|$, proving (2). □

The smallest example, with $n = 4, c = 2$, is

$$C_1 = \{2^{[1]}, 3^{[0]}\}, \quad C_2 = \{1^{[1]}, 3^{[0]}\},$$
$$C_3 = \{4^{[1]}, 1^{[0]}\}, \quad C_4 = \{3^{[1]}, 1^{[0]}\},$$

where brackets denote ages. It works for both $\ell = 1$ and $\ell = 2$. Peer 1 never samples peer 4; its singleton distance is $1/3$. For a realistic illustration, $c = 20, \ell = 5$ is covered whenever even $n \ge 22$.

This counterexample uses a prescribed nonzero-age initialization. It does not apply to mandatory partner removal or duplicate-processing rules that refresh stored ages. The next proposition addresses reachability from zero ages for one parameter choice, while carefully distinguishing pathwise and distributional statements.

Proposition 3 (A frozen state reachable from zero ages)**.** *Under retention, $n = 4, c = 2, \ell = 1$, independent uniform initiators, uniform age-tie resolution, and uniform responder subsets, there is a strongly connected full-cache, zero-age initialization from which a frozen identity state is reached with probability at least 2^{-24}. Consequently almost-sure time-averaged uniform sampling fails for that initialization.*

Proof. Start from

$$C_1 = \{2^0, 4^0\}, \quad C_2 = \{3^0, 4^0\}, \quad C_3 = \{1^0, 2^0\}, \quad C_4 = \{1^0, 2^0\}.$$

Superscripts denote ages. This graph is strongly connected. Execute the following triples of initiator, selected partner, and returned identity:

$$(2,4,1),\ (1,4,2),\ (3,1,2),\ (3,1,4),\ (4,2,1),\ (4,2,3).$$

For $\ell = 1$, an accepted incoming descriptor replaces the single sent entry; a filtered descriptor makes no replacement. Direct substitution in these rules gives

$$C_1 = \{4^1, 3^0\}, \quad C_2 = \{3^1, 4^0\},$$
$$C_3 = \{2^2, 4^1\}, \quad C_4 = \{1^2, 3^1\}.$$

The pairs $(1,4)$ and $(2,3)$ now satisfy the frozen-state condition, with common sets $\{3\}$ and $\{4\}$ respectively. Each of the six specified initiators has probability $1/4$, each specified oldest-partner choice is between two tied entries, and each specified response has probability $1/2$. The exact prefix probability is therefore $(1/16)^6 = 2^{-24}$. On this event, peer 1 never again samples peer 2, so its empirical frequency for that identity tends to zero rather than $1/3$. □

Positive-probability absorption by itself does *not* prove that the unconditional local marginal law fails to approach uniform from the zero-age initialization: different frozen outcomes could balance in their mixture. Theorem 2 directly disproves that marginal-law guarantee when the specified nonzero initial ages are allowed.

4 An availability obstruction under mandatory removal

Now choose initiators independently and uniformly, remove the partner unconditionally, and let an activation of an empty cache do nothing.

Proposition 4 (Persistent failure of full-capacity requests)**.** *Let $c = n-1$ and $1 \le \ell \le c$. For every peer i and $t \ge 1$, even starting from complete caches,*

$$\Pr\{S_i^{(c)}(t) = \perp\} \ge \frac{1}{n}, \qquad \left\| \mathcal{L}(S_i^{(c)}(t)) - \mathrm{Unif}((V \setminus \{i\})_c) \right\|_{\mathrm{TV}} \ge \frac{1}{n}.$$

Proof. If activation t selects i, either its cache is already empty or it removes some partner q. Since q's cache contains no self entry q, its response cannot restore q to i. Thus i has at most $n-2 < c$ entries at the end of that activation. The event that i was selected has probability $1/n$. The distance bound follows by testing the event $\{\perp\}$, which has target probability zero. □

This concerns unconditional $b = c$ requests. It does not establish nonuniformity for smaller samples conditional on availability. Retrying or conditioning changes the sampling model.

5 A positive capacity-one theorem

5.1 Precisely specified variant

Set $c = \ell = 1$, retain the current neighbor when a reply is unusable, and choose the initiator independently and uniformly at each step. With $q = f(p)$ and $r = f(q)$, the complete identity update is

$$T_p f = \begin{cases} f, & r = p, \\ f \text{ with } f(p) \leftarrow r,\ f(q) \leftarrow p, & r \neq p. \end{cases} \tag{3}$$

Initial ages may be arbitrary: the single cached entry is necessarily oldest, so ages have no effect on this identity projection. Let Ω_n be the nonself functions $f : V \to V$ whose underlying undirected graph is connected, and let

$$P(f,g) = \frac{1}{n} \#\{p : T_p f = g\}.$$

Theorem 5 (Convergence of the capacity-one retention variant)**.** *For every $n \ge 2$, $f \in \Omega_n$, and observer i,*

$$\mathcal{L}_f(f_t(i)) \longrightarrow \mathrm{Unif}(V \setminus \{i\}).$$

For $n \ge 4$, with $D = 2n(n-2)$ and $\delta = n^{-D}$,

$$\|\mathcal{L}_f(f_t(i)) - \mathrm{Unif}(V \setminus \{i\})\|_{\mathrm{TV}} \le (1-\delta)^{\lfloor t/D \rfloor}. \tag{4}$$

Consequently $D\lceil n^D \log(1/\varepsilon) \rceil$ activations suffice for distance at most ε. For $n = 3$, $(8/9)^{\lfloor t/2 \rfloor}$ is sufficient. For $n = 2$, the sample is already uniform.

This is an extremely weak quantitative bound and gives no efficient mixing claim. The theorem concerns identity marginals, not stationarity of the complete age process.

5.2 Proof for $n \geq 4$

Lemma 6 (Inverse moves and connectivity)**.** *Ω_n is closed under T_p. Every nontrivial move initiated by p with $q = f(p)$ is inverted by initiating q in the resulting state. In particular P is symmetric.*

Proof. A nontrivial move replaces a path on the three distinct vertices p, q, r by another path on the same vertices, preserving undirected connectivity and avoiding self entries. In the resulting state $g(q) = p, g(p) = r$; initiating q restores $f(q) = r, f(p) = q$. This pairs each marked transition (f, p, g) with its unique reverse marked transition (g, q, f), so the transition counts agree. □

Lemma 7 (Insertion into a cycle)**.** *Every state in Ω_n reaches a directed Hamiltonian cycle in at most $n - 2$ moves.*

Proof. A connected functional digraph has exactly one directed cycle: outgoing paths eventually reach a cycle, and an edge cannot join basins of distinct cycles. The cycle length is at least two. While an off-cycle vertex remains, choose p whose parent $q = f(p)$ lies on the cycle. Initiating p replaces its cycle edge $q \to r$ by $q \to p \to r$, adding one vertex to the cycle. □

Lemma 8 (Adjacent swap)**.** *If a Hamiltonian cycle has consecutive distinct vertices x, a, b, y, the activation word x, x, a, b swaps a, b, leaving every other outgoing edge unchanged.*

Proof. The only changing outgoing edges are

	$f(x)$	$f(a)$	$f(b)$
initial	a	b	y
x	b	x	y
x, x	y	x	x
x, x, a	a	y	x
x, x, a, b	b	y	a

All moves are valid for $n \geq 4$. At $n = 4$ the temporary residual cycle $x \leftrightarrow y$ is allowed. □

Lemma 9 (Diameter)**.** *For $n \geq 4$, the transition graph on Ω_n is connected with diameter at most $D = 2n(n - 2)$.*

Proof. Reach a Hamiltonian cycle from each of two states in at most $n - 2$ steps. Anchor the cyclic notation at one fixed vertex. The other $n-1$ vertices form a linear ordering; any two orders are related by at most $(n - 1)(n - 2)/2$ adjacent swaps. Each costs four moves by Lemma 8. Reverse the second insertion path using Lemma 6. The total is at most $2(n-2)+2(n-1)(n-2) = 2n(n - 2)$. □

Proof of Theorem 5 for $n \geq 4$. The uniform law π_n on Ω_n is stationary because P is symmetric. Choose the state $z(1) = 2, z(2) = 1, z(j) = 1$ for $j \geq 3$. Initiating 1 at z is a self-loop. Every state reaches z along a path of at most D moves. Pad shorter paths with that self-loop to exactly D activations. The prescribed sequence has probability $n^{-D} = \delta$, so

$$P^D(f, \cdot) \geq \delta\, \delta_z(\cdot) \quad (f \in \Omega_n).$$

Writing $P^D(f, \cdot) = \delta\, \delta_z + (1 - \delta)R(f, \cdot)$ yields

$$\|\mu P^D - \nu P^D\|_{\mathrm{TV}} \leq (1 - \delta)\|\mu - \nu\|_{\mathrm{TV}}$$

for any probability laws μ, ν. Iterate with $\nu = \pi_n$, then use total-variation contraction for the remaining steps:

$$\|P^t(f, \cdot) - \pi_n\|_{\mathrm{TV}} \le (1 - \delta)^{\lfloor t/D \rfloor}.$$

Relabelling bijects Ω_n onto itself and preserves π_n. Permutations fixing i act transitively on the other identities, so $\pi_n\{f : f(i) = j\} = 1/(n-1)$ for each $j \neq i$. Projection gives (4). The time bound follows from $(1 - \delta)^k \le e^{-\delta k}$. □

5.3 Small network sizes

For $n = 2$ the sole state is the reciprocal pair and its local target distribution has one element. For $n = 3$ the identity chain has two four-state classes. One consists of

$$H = (2, 3, 1), \quad L_1 = (3, 1, 1), \quad L_2 = (2, 1, 2), \quad L_3 = (3, 3, 2).$$

The other is obtained by relabelling a reversed triangle. The center H moves to each leaf with probability $1/3$. Each leaf returns to H with probability $1/3$ and stays with probability $2/3$. These classes contain all eight nonself maps. Uniform law on each class is stationary, and every observer points to either other identity in exactly two of its four states.

Choose a leaf $L_\star$ in the appropriate class. From every state, $P^2(f, L_\star) \ge 1/9$: from another leaf go through H; from H use the target leaf's self-loop; from the target leaf use two self-loops. The same minorization argument proves the stated bound. This also shows why irreducibility of the full identity space is stronger than necessary for local uniformity.

6 A general symmetry reduction, not a mixing proof

Lemma 10 (Symmetrized reference process)**.** *Let a Markov transition kernel commute with peer relabelling, and suppose the sample extraction rule is equivariant and always defined. Fix initial state x, observer i, and the group H_i of permutations fixing i. Start a reference process Y_t from*

$$\bar{\mu}_x = \frac{1}{|H_i|} \sum_{\sigma \in H_i} \delta_{\sigma x}.$$

Then $S_i^{(b)}(Y_t)$ is exactly uniform on the ordered distinct b-tuples of other peers at every time. For any coupling,

$$\|\mathcal{L}_x(S_i^{(b)}(X_t)) - \mathrm{Unif}((V \setminus \{i\})_b)\|_{\mathrm{TV}} \le \Pr\{S_i^{(b)}(X_t) \neq S_i^{(b)}(Y_t)\}.$$

Proof. The initial law of Y is invariant under H_i. Equivariance preserves this invariance at every time and transfers it to the sample law. Since H_i acts transitively on the output tuples, that law must be uniform. The remaining claim is the coupling inequality. □

This lemma requires no finite state space, stationary age law, or full-state irreducibility. If samples can be unavailable, the uniformity conclusion holds only conditional on availability. The lemma provides no forgetting bound by itself. In Theorem 2, its reference law is uniform while the original local law remains permanently nonuniform, so the required disagreement probability cannot tend to zero.

7 Established results and remaining problem

The fixed-period blanket convergence claim is false by Theorem 1. Retention with duplicate-age preservation also admits permanently frozen, strongly connected multi-entry overlays, even

with random scheduling, by Theorem 2. Mandatory removal avoids that construction, but a full-capacity sampling theorem must address Proposition 4.

Theorem 5 establishes local uniformity only for the specified capacity-one retention variant. It does not establish multi-entry, oldest-first CYCLON convergence. For that question, one must choose a precise implementation and a valid sampling target, exclude or escape the obstructions above, and prove a new quantitative forgetting argument for its cache-and-age dynamics. Neither eviction counts alone nor the symmetry lemma supplies that argument. No polylogarithmic mixing claim is established here.

8 Finite verification

The accompanying `verify.py` checks the periodic trace including ages, every possible initiating/subset choice in a small frozen-twin example, the zero-age absorption certificate, the frozen construction's connectivity, the availability argument on small cache states, and the capacity-one identity chain exhaustively for $2 \leq n \leq 6$. It checks inverse moves, communicating classes, stationary local counts, and Hamiltonian adjacent swaps. These are corroborating checks; the proofs establish arbitrary-n statements independently of enumeration.

References

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